

ABHINAV A

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Professional Summary

Python Full-Stack Developer with expertise in building scalable, production-ready web applications using Django, FastAPI, PostgreSQL, Redis, Docker, AWS, and React.js. Experienced in designing high-performance REST APIs, geospatial services, real-time systems with WebSockets, secure authentication, AI-powered features, and cloud-native architectures. Passionate about solving complex backend challenges through clean architecture, scalable system design, and efficient database optimization.

Technical Skills

Languages	Python, JavaScript, SQL, HTML, CSS
Frontend	React.js, Vite, Tailwind CSS, Leaflet, Recharts, Axios
Backend	Django, DRF, FastAPI, Celery, Celery Beat, Daphne, Django Channels, WebSockets
Database	PostgreSQL, PostGIS, pgvector, Redis, GeoDjango
AI & ML	Groq/Llama 3.3, SentenceTransformers, pgvector (RAG)
Cloud & DevOps	Docker, AWS EC2, AWS RDS, GitHub Actions, Nginx, Git
Tools	Swagger (drf-spectacular), Google OAuth

Experience

Full-Stack Developer Intern

Sep 2025 – Present

Bridgeon Solutions, Kozhikode

- Engineer RESTful APIs with Django and DRF for live production modules, enforcing JWT auth, role-based access, and input validation across multiple client-facing endpoints.
- Eliminate N+1 query patterns using `select_related` and `prefetch_related`, cutting ORM query count on critical endpoints and improving response times.
- Define and maintain API contracts between backend and frontend teams, preventing interface mismatches across delivery cycles.

Diploma Trainee Engineer

Dec 2021 – Dec 2022

Bajaj Auto Pvt Ltd, Pune

- Tracked production line metrics and diagnosed mechanical faults under time-critical conditions in a high-volume manufacturing environment.

Projects

EcoCharge – Smart EV Charging & Trip Planning Platform

ecocharge-nine.vercel.app | [Github](#)

Django, FastAPI, PostgreSQL/PostGIS, pgvector, Redis, Celery, Django Channels, Daphne, Docker, React, Vite, Tailwind CSS, Leaflet, Recharts

- Built a concurrent-safe slot reservation engine using `select_for_update()` inside atomic transactions with overlapping time-window detection, integrated Razorpay for automated payment capture, refund processing, and signature verification.
- Developed a FastAPI microservice with a multi-tool LLM agent (Groq/Llama 3.3) featuring RAG via pgvector + SentenceTransformers, Redis-backed user memory, prompt injection guards, PII/Aadhaar redaction, and SSE-streaming responses with a quality verification loop.
- Designed a segment-by-segment energy-aware route planner using vehicle-specific charging curves, OSRM routing with Redis-cached distance matrices, and concurrent weather fetching via ThreadPoolExecutor.
- Built a data pipeline ingesting from 4 sources (Open Charge Map API, Kaggle, HuggingFace, and synthetic highway generation via OSRM) with India boundary validation, PostGIS spatial indexing, and deduplication across 400+ cities and 55+ highway corridors.
- Built the React (Vite + Tailwind CSS) frontend with Leaflet route/station maps, Recharts analytics dashboards, and Google OAuth login; containerized 6 services (Django, FastAPI, PostgreSQL/PostGIS, Redis, Celery, Nginx) with GitHub Actions CI/CD deploying to AWS EC2 via GHCR, Let's Encrypt SSL, and real-time slot updates via Daphne-served Django Channels + WebSockets.

CricGear – E-Commerce Platform

crick-gear-ecommerce.vercel.app | [Github](#)

Django, DRF, PostgreSQL, React.js, AWS EC2, AWS RDS, JWT

- Architected and deployed a full e-commerce backend on AWS EC2 with PostgreSQL on RDS, configuring all infrastructure including security groups and environment settings independently.
- Designed 15+ REST endpoints with JWT authentication, role-based access control, and field-level validation, verified end-to-end via Postman before frontend handoff.
- Integrated React.js frontend with Django APIs for product listing, cart, and order placement using React Hooks and Axios, shipped as a working live product.
- Handled the full deployment lifecycle including EC2 provisioning, RDS setup, and environment hardening, gaining end-to-end production AWS experience independently.

Education

Bachelor of Computer Applications (BCA)

2024 – 2027

Manipal University Jaipur

Diploma in Mechanical Engineering (Lateral Entry)

2019 – 2021

Kerala Govt Polytechnic College

CGPA: 8.57 / 10